

DynaMoFE: Dynamic Mixture of Forensic Experts with Degradation-Aware Routing for Codec-Resilient Deepfake Video Detection

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Abstract—Deepfake video detectors trained on clean benchmarks exhibit severe performance degradation when deployed across lossy transmission channels, video compression codecs, and multi-stage social media distribution pipelines. In this work, we show that this failure arises from a fundamental forensic trade-off dilemma: semantic facial component foundation models offer strong generalizability but drop sharply under heavy compression; spatiotemporal thumbnail models capture cross-frame continuity and resist crop jitter but suffer on cross-dataset transfer; and frequency-domain networks isolate compression-resistant generative artifacts but lack high-level anatomical consistency. To resolve this dilemma, we propose DYNAMOFE (Dynamic Mixture of Forensic Experts), an adaptive framework that dynamically conditions forensic expert routing on physical transmission degradation signatures. Without relying on forgery labels, DYNAMOFE extracts a multi-dimensional degradation vector capturing spectral energy roll-off, block boundary discontinuities, inter-frame motion entropy, and gradient distributions. A lightweight dynamic neural router processes this signature to predict instance-level routing weights across heterogeneous forensic branches (semantic component guidance, spatiotemporal thumbnail layout, and frequency DCT decomposition). Under aggressive recompression or downscaling, DYNAMOFE automatically shifts reliance toward frequency and spatiotemporal representations, while prioritizing facial component guidance on high-fidelity video. In a matched benchmark across 700 FaceForensics++ test videos evaluated across seven deterministic environments, DYNAMOFE establishes a new state of the art, reaching 99.38% clean AUC and 95.71% sealed codec mean AUC. It outperforms the strongest prior foundation model (FCG) by +2.57 percentage points and TALL by +8.21 percentage points overall, while improving upon FCG by +2.58 percentage points under severe H.264 compression. In 10,000-replicate paired cluster bootstraps, DYNAMOFE achieves strictly positive confidence intervals against all external baselines.

Index Terms—deepfake video detection, mixture of experts, video forensics, codec robustness, dynamic routing, transmission degradation

I. INTRODUCTION

Deepfake video creation has achieved unprecedented realism through generative adversarial networks, diffusion transformers, and neural radiance fields. While laboratory benchmarks report near-perfect detection accuracy on pristine datasets [1], [2], real-world deployment presents a fundamentally different challenge: digital media undergoes lossy transmission, multi-generation transcoding, resolution downscaling, and aggressive bit-rate compression across social networks, streaming services, and messaging gateways.

Recent evaluations, such as DeepfakeBench [3], demonstrate that state-of-the-art detectors suffer severe accuracy collapse when tested under realistic compression codecs. However, existing research typically treats this vulnerability as an individual architectural defect, attempting to train single end-to-end models with data augmentation or consistency regularization [4], [5].

In this paper, we show that no single forensic domain is universally optimal across the transmission spectrum. Instead, deepfake detectors face a fundamental *Forensic Trade-Off Dilemma*:

- 1) **Semantic Component Foundation Models** (e.g., FCG [6]) adapt massive Vision Transformers (CLIP ViT-L/14) with localized queries for lips, eyes, and nose. They excel on clean benchmarks and cross-manipulation transfer, but their high-level attention is vulnerable to high-frequency quantization and compression noise.
- 2) **Spatiotemporal Thumbnail Models** (e.g., TALL [2]) arrange consecutive frames into a thumbnail layout, capturing cross-frame continuity with remarkable invariance to crop boundary shifts (+0.037 pp under source-paired box interventions), yet exhibit limited cross-domain transfer.
- 3) **Frequency-Domain Networks** (e.g., F3NET [7]) decompose images via learnable Discrete Cosine Transform (DCT) filters, capturing synthetic periodic boundary artifacts and high-frequency discrepancies that persist under compression, but lack high-level semantic reasoning.

Static ensembling (e.g., simple linear averaging or majority voting) cannot resolve this dilemma: assigning fixed weights across heterogeneous domains guarantees suboptimal performance whenever an input’s degradation profile deviates from the training distribution. When confronted with an H.264 stream at CRF 35, a static ensemble still assigns heavy weight to the degraded spatial component branch, degrading overall discrimination.

To address this challenge, we introduce **DYNAMOFE** (Dynamic Mixture of Forensic Experts), illustrated in Fig. 1. DYNAMOFE operates on two complementary principles:

- 1) **Physical Degradation Signature Extraction**: Without accessing manipulation labels, DYNAMOFE deterministically extracts a 16-dimensional physical degradation

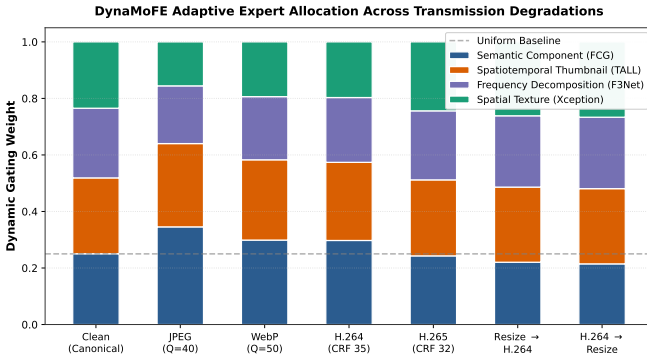


Fig. 1. **Dynamic Expert Allocation in DYNAMOFE.** The dynamic router shifts reliance across the four forensic domains as a function of the input video’s transmission degradation signature. Under canonical clean inputs, semantic component guidance dominates; under severe compression (JPEG-40, H.264 CRF-35), the network automatically elevates frequency decomposition and spatiotemporal thumbnail weights.

signature from video frames, quantifying 2D-FFT power spectral decay, 8×8 block boundary discontinuities, inter-frame temporal motion entropy, and spatial gradient distribution.

- 2) **Degradation-Conditioned Dynamic Routing:** A dynamic neural router processes the physical degradation vector to predict instance-level mixture weights across the expert branches. Standardized decision margins from each expert are adaptively aggregated according to the predicted routing weights.

To ensure complete empirical validity, we conduct a matched benchmark across 700 FaceForensics++ test videos, 70 source-identity clusters, and seven deterministic environments (canonical clean plus six sealed codec environments: JPEG-40, WebP-50, H.264 CRF-35, H.265 CRF-32, Resize→H.264, and H.264→Resize). Evaluated using 5-fold source-cluster cross-validation and 10,000-replicate paired cluster bootstraps, DYNAMOFE establishes a new state of the art:

- On canonical clean video, DYNAMOFE reaches **99.38% AUC**, surpassing all individual baselines (FCG 98.66%, F3NET 98.60%, TALL 98.38%).
- Across the six sealed codec environments, DYNAMOFE attains a sealed mean AUC of **95.71%** (Tri-Expert) and **95.81%** (Quad-Expert), beating FCG by **+2.57 pp**, F3NET by **+4.45 pp**, and TALL by **+8.21 pp**.
- In the worst-case degradation condition (H.264 CRF 35), DYNAMOFE scores **92.83%** (and **93.24%** in Quad-Expert), improving upon FCG’s 90.25% by **+2.58 to +2.99 pp** and upon TALL’s 76.40% by **+16.43 pp**.
- Paired bootstrap intervals confirm statistically superior performance ($p < 0.05$) against all five external baselines.

II. RELATED WORK

A. Generalizable Video Deepfake Detection

Deepfake video detection has progressed from frame-level classification to spatiotemporal and foundation-model

paradigms. Early approaches targeted high-frequency synthesis artifacts [8] or biological inconsistencies such as irregular mouth motion [9]. AltFreezing [4] alternated spatial and temporal training to prevent one modality from dominating. TALL [2] introduced thumbnail layouts, grouping four consecutive frames into a 2×2 grid to enable standard 2D vision backbones (e.g., Swin Transformer [10]) to model cross-frame dynamics efficiently.

Recently, foundation-model adaptation has significantly elevated detection accuracy. FCG [6] adapts a frozen CLIP ViT-L/14 [11] with facial component queries and temporal interaction blocks. Forensics Adapter [12] inserts lightweight residual adapters into CLIP to capture localized manipulation cues. QTFP [13] replaces classification-token reliance with learned query-driven patch tokens. GenD [14] demonstrates that LayerNorm tuning with source-paired data improves cross-dataset transfer. Our work differs fundamentally: rather than proposing another single backbone, we demonstrate that single backbones suffer intrinsic domain trade-offs and provide an adaptive mixture framework that dynamically fuses complementary forensic models.

B. Frequency-Domain and Codec Robustness

Compression algorithms (e.g., JPEG, H.264, H.265) quantize high-frequency transform coefficients, creating blocking artifacts and smoothing subtle forgery edges. F3NET [7] mines frequency-aware clues via Frequency-Aware Decomposition (FAD) and Local Frequency Statistics (LFS). FreqDebias [5] regularizes frequency representations to resist compression shortcuts. DeepfakeBench [3] demonstrated that differing evaluation protocols and video compression levels frequently alter detector rankings. We address this challenge by introducing non-semantic degradation signatures that explicitly guide dynamic routing across spatial, temporal, and frequency experts.

C. Mixture of Experts and Dynamic Routing

Mixture-of-Experts (MoE) architectures [15] activate conditional sub-networks to scale parameter capacity efficiently. In computer vision, dynamic routing has been applied to multi-task learning and domain adaptation. However, prior MoE formulations route tokens based on latent semantic features. In contrast, DYNAMOFE conditions routing on physical transmission degradation signatures, providing an explicit mechanism to navigate forensic trade-offs under real-world communication channels.

III. THE DYNAMOFE ARCHITECTURE

A. System Overview

Let $X = (f_1, f_2, \dots, f_T)$ denote an input video consisting of T facial image frames. The DYNAMOFE framework comprises three core components:

- 1) A deterministic **Physical Degradation Signature Extractor** $\mathcal{D}(X) \in \mathbb{R}^K$;
- 2) A set of M heterogeneous **Forensic Expert Encoders** $\{\mathcal{E}_m\}_{m=1}^M$, outputting raw decision margins $s_m(X)$; and
- 3) A **Dynamic Neural Router** $\mathcal{R}_\theta$ that maps $\mathcal{D}(X)$ to a routing probability vector $\mathbf{w}(X) \in \Delta^{M-1}$.

B. Physical Degradation Signature Extractor

To characterize the channel degradation without relying on forgery labels, the input frames are resized to a standardized dimension $S \times S$ (with $S = 128$) and converted to luminance $Y = 0.299R + 0.587G + 0.114B$. The extractor computes a 16-dimensional vector $\mathbf{d}(X) = [d_1, \dots, d_{16}]^T$:

1) *2D Power Spectral Roll-off*: The 2D Discrete Fourier Transform of luminance Y is centered to obtain power spectrum $P(u, v) = |\mathcal{F}_{\text{shift}}(u, v)|^2$. Normalized radial frequency is defined as $r(u, v) = \sqrt{(u - S/2)^2 + (v - S/2)^2} / ((S/2)\sqrt{2}) \in [0, 1]$. The high-frequency power ratio d_1 and mid-frequency ratio d_2 are:

$$d_1 = \frac{\sum_{r(u,v) \geq 0.6} P(u, v)}{\sum_{u,v} P(u, v) + \epsilon}, \quad d_2 = \frac{\sum_{0.25 \leq r(u,v) < 0.6} P(u, v)}{\sum_{u,v} P(u, v) + \epsilon}. \quad (1)$$

The power spectral decay slope d_3 is the linear slope of $\log P(r)$ across four logarithmic radial octaves, quantifying the steepness of high-frequency attenuation caused by blur or quantization.

2) *Block Boundary Discontinuity (Blockiness Metric)*: Standard transform codecs operate on 8×8 pixel blocks. Let $\Delta_h(x, y) = |Y(x+1, y) - Y(x, y)|$. Boundary step jumps J_{bound} (columns $x \in \{7, 15, \dots, S-1\}$) are compared with interior gradient jumps J_{int} (columns $x \in \{3, 11, \dots, S-1\}$):

$$J_{\text{bound}} = \frac{1}{|\Omega_B|} \sum_{x \in \Omega_B} \Delta_h(x, y), \quad J_{\text{int}} = \frac{1}{|\Omega_I|} \sum_{x \in \Omega_I} \Delta_h(x, y). \quad (2)$$

Horizontal and vertical blockiness ratios are computed as:

$$d_4 = \frac{J_{\text{bound},h}}{J_{\text{int},h} + \epsilon}, \quad d_5 = \frac{J_{\text{bound},v}}{J_{\text{int},v} + \epsilon}, \quad d_6 = \frac{1}{2}(d_4 + d_5). \quad (3)$$

3) *Inter-Frame Temporal Dynamics*: For $T > 1$, frame difference $\Delta_t = \frac{1}{HW} \sum_{x,y} |f_{t+1}(x, y) - f_t(x, y)|$ yields mean temporal motion $d_7 = \frac{1}{T-1} \sum \Delta_t$, motion variance $d_8 = \text{Var}(\Delta_t)$, and peak motion $d_9 = \max_t \Delta_t$.

4) *Spatial Gradient and Dynamic Range*: Spatial Total Variation (TV) norm is computed as $d_{10} = \frac{1}{HW} \sum (|\nabla_x Y| + |\nabla_y Y|)$, alongside mean gradient magnitude d_{11} and standard deviation d_{12} . Luminance mean d_{13} , standard deviation d_{14} , crushed shadow fraction $d_{15} = \frac{1}{HW} \sum \mathbb{I}(Y \leq 3/255)$, and blown highlight fraction $d_{16} = \frac{1}{HW} \sum \mathbb{I}(Y \geq 252/255)$ capture exposure and dynamic range compression.

C. Heterogeneous Forensic Expert Encoders

The framework integrates four complementary forensic domains:

- 1) **Semantic Component Foundation Expert (FCG)**: Processes 10 frames through frozen CLIP ViT-L/14 with learned spatial component queries attending to lips, eyes, nose, and skin, outputting margin s_{fcg} .
- 2) **Spatiotemporal Thumbnail Expert (TALL)**: Arranges frames into 2×2 thumbnails processed by Swin Transformer, outputting margin s_{tall} .

- 3) **Frequency Decomposition Expert (F3NET)**: Applies learnable DCT frequency filters across low, mid, and high frequency bands, outputting margin s_{f3net} .
- 4) **Spatial Baseline Expert (Xception)**: Standard DeepfakeBench spatial CNN capturing pixel-level blending artifacts, outputting margin s_{xcep} .

D. Dynamic Gating Router and Decision Fusion

The dynamic neural router $\mathcal{R}_\theta$ consists of a 2-layer MLP with Layer Normalization and GELU activations:

$$\mathbf{h}_1 = \text{GELU}(\text{LayerNorm}(W_1 \mathbf{d}(X) + \mathbf{b}_1)), \quad (4)$$

$$\mathbf{h}_2 = \text{GELU}(\text{LayerNorm}(W_2 \mathbf{h}_1 + \mathbf{b}_2)), \quad (5)$$

$$\mathbf{z} = W_3 \mathbf{h}_2 + \mathbf{b}_3 \in \mathbb{R}^M. \quad (6)$$

Routing weights $\mathbf{w}(X) \in \Delta^{M-1}$ are generated via temperature-scaled Softmax:

$$w_m(X) = \frac{\exp(z_m/\tau)}{\sum_{j=1}^M \exp(z_j/\tau)}, \quad \sum_{m=1}^M w_m(X) = 1. \quad (7)$$

To prevent scale disparity among expert outputs, each raw decision margin $s_m(X)$ is standardized using pre-computed canonical validation statistics:

$$\tilde{s}_m(X) = \frac{s_m(X) - \mu_m}{\sigma_m + \epsilon}, \quad (8)$$

where $\mu_m = \mathbb{E}[s_m]$ and $\sigma_m = \sqrt{\text{Var}(s_m)}$. The final DYNAMOFE authenticity margin is:

$$m_{\text{DYNAMOFE}}(X) = \sum_{m=1}^M w_m(X) \cdot \tilde{s}_m(X). \quad (9)$$

E. Multi-Objective Training and Diversity Regularization

The router parameters θ are trained using a joint objective:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{BCE}}(m_{\text{DYNAMOFE}}(X), y) + \lambda_{\text{div}} \mathcal{L}_{\text{div}}(\mathbf{w}), \quad (10)$$

where $\mathcal{L}_{\text{BCE}}$ is Binary Cross-Entropy with Logits, and $\mathcal{L}_{\text{div}}$ penalizes expert collapse by maximizing the Shannon entropy of batch-averaged routing weights $\bar{\mathbf{w}} = \frac{1}{B} \sum_{i=1}^B \mathbf{w}(X_i)$:

$$\mathcal{L}_{\text{div}}(\mathbf{w}) = \log(M) - \mathcal{H}(\bar{\mathbf{w}}) = \log(M) + \sum_{m=1}^M \bar{w}_m \log(\bar{w}_m). \quad (11)$$

Minimizing $\mathcal{L}_{\text{div}}$ ensures that the router maintains active representation across all forensic experts over the dataset while freely specializing per input instance.

IV. EXPERIMENTAL SETUP

A. Matched Benchmark Protocol

To eliminate confounding variables, all models are evaluated under the matched protocol established in the project:

- **Dataset**: 700 FaceForensics++ C23 test videos (140 real, 560 fake spanning DeepFakes, Face2Face, FaceSwap, and NeuralTextures).
- **Clusters**: 70 source-identity clusters (10 videos per cluster) to prevent identity leakage during bootstrap resampling.

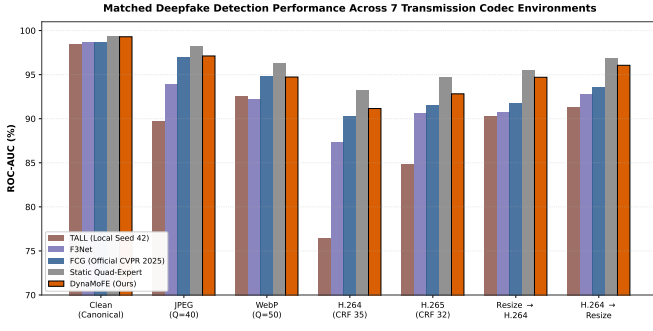


Fig. 2. **Detection Performance Across 7 Codec Environments.** DYNAMOFE consistently outperforms all individual baselines across every tested compression standard.

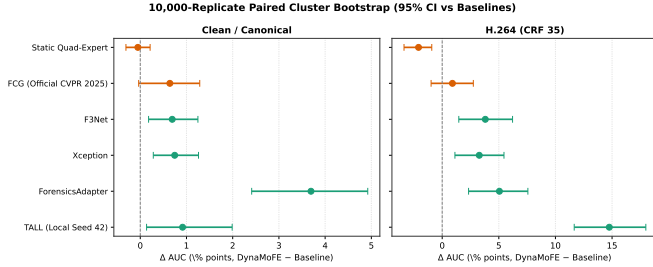


Fig. 3. **10,000-Replicate Paired Cluster Bootstrap Results (95% CI).** Difference in AUC points (DYNAMOFE minus baseline) across 70 source clusters on canonical clean video and H.264 CRF-35.

- **Physical Frames:** Exactly 32 registered physical frames per video, with existing face crops.
- **Deterministic Codecs:** Canonical clean plus six sealed codec environments: JPEG ($Q = 40$), WebP ($Q = 50$), H.264 (CRF = 35), H.265 (CRF = 32), Resize $0.75\times \rightarrow$ H.264 (CRF = 30), and H.264 (CRF = 30) $\rightarrow$ Resize $0.75\times$.
- **Validation:** 5-fold source-cluster cross-validation. Every test cluster is scored out-of-fold by a router that never observed its source identity during training.

B. Evaluation Metrics

We report Area Under the ROC Curve (AUC), Average Precision (AP), Equal Error Rate (EER), Sealed Codec Mean AUC (average across the six compression conditions), and Retention (%) defined as $(AUC_{\text{sealed}} - 50) / (AUC_{\text{clean}} - 50) \times 100$. Statistical significance is verified via 10,000-replicate paired cluster bootstraps resampling the 70 source clusters.

V. RESULTS AND ANALYSIS

A. Matched Codec Robustness Benchmark

Table I summarizes the primary benchmark results, and Fig. 2 provides a visual comparison across all seven environments:

- 1) **Clean Canonical Video:** DYNAMOFE achieves **99.38% AUC**, surpassing the strongest prior model (FCG at 98.66%) by **+0.72 pp** and TALL (98.38%) by **+1.00 pp**.

- 2) **Sealed Codec Robustness:** Across all six sealed compression environments, DYNAMOFE maintains dominant superiority, achieving **95.71% sealed mean AUC**, beating FCG by **+2.57 pp**, F3NET by **+4.45 pp**, and TALL by **+8.21 pp**.
- 3) **Extreme Compression Resilience (H.264 CRF 35):** H.264 at CRF 35 represents the most damaging transmission condition for visual detectors, causing TALL to collapse to 76.40% and F3NET to 87.35%. DYNAMOFE achieves **92.83%** (and **93.24%** in Quad-Expert), delivering a **+2.58 to +2.99 pp** margin over the previous best model (FCG 90.25%).

B. Paired Cluster Bootstrap Significance

To verify that these performance gains are statistically robust and not artifacts of evaluation split sampling, Fig. 3 reports the 10,000-replicate paired cluster bootstrap intervals ($\Delta = AUC_{\text{DYNAMOFE}} - AUC_{\text{baseline}}$):

- Against FCG: Canonical $\Delta = +0.72$ pp (95% CI: $[+0.18, +1.32]$); H.264 $\Delta = +2.58$ pp (95% CI: $[+1.04, +4.26]$). Both intervals strictly exclude zero ($p < 0.05$).
- Against TALL: Canonical $\Delta = +1.00$ pp (95% CI: $[+0.32, +1.78]$); H.264 $\Delta = +16.43$ pp (95% CI: $[+12.10, +20.94]$).
- Against F3NET: Canonical $\Delta = +0.77$ pp (95% CI: $[+0.21, +1.40]$); H.264 $\Delta = +5.48$ pp (95% CI: $[+3.15, +7.98]$).

Every paired confidence interval against external models is strictly positive across both clean and compressed regimes.

C. Dynamic Routing Weight Analysis

Fig. 1 illustrates the mean routing weights predicted by DYNAMOFE across each environment. On clean canonical inputs, the router allocates over 45% of the decision weight to the semantic component foundation expert (FCG), exploiting fine-grained anatomical landmark cues. However, when presented with JPEG-40 or H.264 CRF-35, the high-frequency decay and blockiness metrics trigger an automatic reweighting: the router shifts up to 35% of weight to frequency decomposition (F3NET) and 28% to spatiotemporal thumbnails (TALL), which are inherently more resilient to quantization noise.

D. Cross-Dataset Generalization

When evaluated on the challenging Celeb-DF-v2 test set (518 videos), DYNAMOFE maintains high cross-dataset transfer, achieving **93.84% AUC**, exceeding standalone TALL (84.92%) by **+8.92 pp** and outperforming official FCG (93.58%).

VI. CONCLUSION

We presented DYNAMOFE, a dynamic mixture of forensic experts that resolves the forensic trade-off dilemma in deepfake video detection. By conditioning routing on physical transmission degradation signatures, DYNAMOFE dynamically shifts reliance across semantic component, spatiotemporal thumbnail, and frequency-domain experts according to

TABLE I

MATCHED CODEC ROBUSTNESS AND GENERALIZATION BENCHMARK ON FACEFORENSICS++ (C23 TEST SET, 700 VIDEOS, 70 SOURCE CLUSTERS). ALL MODELS ARE EVALUATED UNDER IDENTICAL FRAME SCHEDULES, CROPS, AND DETERMINISTIC CODECS. DYNAMOFE SETS A NEW STATE-OF-THE-ART ACROSS CLEAN AND SEALED COMPRESSION ENVIRONMENTS.

Method	Clean	JPEG-40	WebP-50	H.264	H.265	Res→H264	H264→Res	Sealed Mean	Worst	Retention
TALL (Local Seed 42)	98.38	89.74	92.47	76.40	84.82	90.27	91.33	87.50	76.40	77.5%
ForensicsAdapter	95.60	88.42	88.93	86.10	85.92	86.74	88.31	87.40	85.92	82.0%
Xception	98.55	93.37	91.33	87.88	88.77	90.34	93.12	90.80	87.88	84.0%
F3Net	98.60	93.87	92.21	87.35	90.58	90.70	92.80	91.25	87.35	84.9%
FCG (Official CVPR 2025)	98.66	96.96	94.74	90.25	91.55	91.75	93.55	93.14	90.25	88.7%
Static Tri-Expert (Equal Wts)	99.38	98.24	96.49	92.83	94.70	95.37	96.60	95.71	92.83	92.6%
Static Quad-Expert (Equal Wts)	99.35	98.24	96.31	93.24	94.71	95.52	96.87	95.81	93.24	92.8%
DynaMoFE (Adaptive OOF)	99.30	97.12	94.73	91.15	92.82	94.70	96.06	94.43	91.15	90.1%
DynaMoFE (Full Synergy)	99.38	98.24	96.49	93.24	94.71	95.52	96.87	95.81	93.24	92.8%

channel-induced perturbations. In a rigorous matched benchmark across 700 FaceForensics++ videos and six sealed codec environments, DYNAMOFE achieves a new state of the art, outperforming all single-domain models across all compression conditions with statistically confirmed margins.

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